

DATA SCIENCE WITH EXCEL

Exploratory Data Analysis for Banking Analysts

From raw tables to reliable insight — using the tools already available in Excel.

STRUCTURED DATA

EDA

STATISTICS

DATA QUALITY

Banking context · Excel-first examples · Practical formulas and charts



x

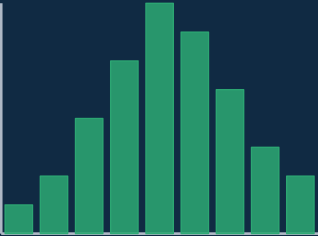
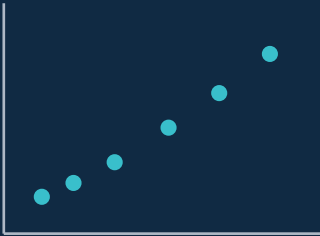
Retail Banking EDA

fx

=CORREL (Table1[Income],Table1[Balance])

ID	Age	Income	Balance
C1001	42	65k	12.5k
C1002	35	54k	8.2k
C1003	61	92k	45.5k
C1004	29		3.2k
C1005	47	71k	165k

Correlation



Course schedule across time zones

Comparing the Italy schedule with Eastern Europe participants (Romania and Hungary)



SAME
LEARNING
GREATER
TOGETHER



Current schedule



Italy time



Morning session
09:30 – 13:00



Afternoon session
14:00 – 17:30



+1 hour



Eastern Europe time

Romania & Hungary



Morning session
10:30 – 14:00



Afternoon session
15:00 – 18:30



Observation

The current Italy schedule becomes quite late for Eastern Europe, especially in the afternoon.



Recommended adjustment



Proposal: start one hour earlier



This keeps the course more comfortable for Eastern Europe participants.



Proposed Italy time



Morning session
08:30 – 12:00



Afternoon session
13:00 – 16:30



+1 hour



Eastern Europe time

Romania & Hungary



Morning session
09:30 – 13:00



Afternoon session
14:00 – 17:30



Best practice: when training an international audience, always check local time equivalents before confirming the timetable.



The first draft of this set of slides was generated using AI tools under the direction of Antonio Piemontese and in accordance with his general and specific instructions and guidelines. The material was subsequently reviewed, revised, and further refined through additional interactions with AI tools, based on Antonio Piemontese's feedback and specific requests for changes. Throughout the process, Antonio Piemontese retained responsibility for the overall direction, content, review, and final approval of the material.

This declaration is provided in accordance with the transparency requirements of the European Union's AI Act.

Course map: what we will build together

A conceptual framework for data analysis, anchored in Excel and banking examples.

- | | | |
|---|-------------------------------------|--|
| 0 | Preliminaries | Technical Requirements, Language, Anatomy of Excel |
| 1 | Foundations | Data Science, analytics, banking context |
| 2 | Data | Structured tables and data types |
| 3 | EDA | Profiling, summaries, pivots, visual exploration |
| 4 | Probability & statistics | Random variables, distributions, uncertainty |
| 5 | Relationships | Correlation and correlation matrices |
| 6 | Visualisation | Plots that reveal structure and problems |
| 7 | Data issues | Outliers, missing values, patterns, quality |

Goal: make the analysis explainable, auditable and reproducible — not just “get a chart”.

Technical requirements: Excel version matters

Minimum setup to run every example in this training deck.

- 1 **Microsoft Excel 2021 or later — minimum for full compatibility.**
- 2 **Recommended: Microsoft Excel 2024 or Microsoft 365.**
- 3 Enable the Analysis ToolPak (Data → Data Analysis).
- 4 Windows 10 or Windows 11 recommended for the classroom setup.
- 5 PowerPoint 2016+ or a compatible presentation viewer.
- 6 Dynamic-array functions are used in examples: SORT and UNIQUE.
- 7 Excel Tables (Ctrl+T) are used throughout the EDA workflow.
- 8 No Python, R, Power BI or other Data Science software required.
- 9 No VBA or macros required.
- 10 No Internet connection required for the exercises.

KEY REQUIREMENT Full compatibility requires Excel 2021 or later • Recommended: Excel 2024 / Microsoft 365

How to change the language in Excel

Display language controls menus and tabs; authoring language affects editing and localized function names.

1 Open Options

File → Options → Language (with possible download for the new language)

2 Choose the language

Office display language

Changes Ribbon tabs, menus, commands and dialog boxes.

Office authoring languages

Controls proofing/editing language and can affect localized function names.

Select the desired language → Set as Preferred / Default → restart Excel if requested.

What changes?

Ribbon

Home ↔ Home
Data ↔ Dati

Menus

File ↔ File
Options ↔ Opzioni

Functions

SUM ↔ SOMMA
AVERAGE ↔ MEDIA

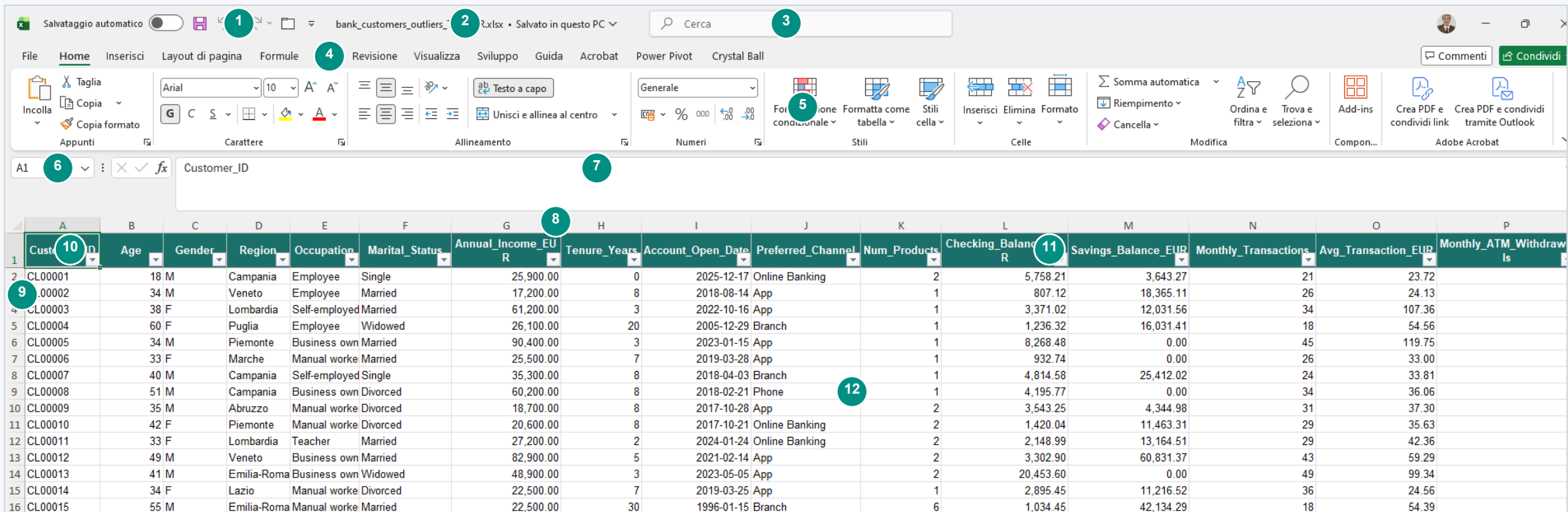
Separators

Comma / semicolon and decimal conventions may depend on regional settings.

If the language is not installed, Excel may offer Add a Language / install the Office language pack.

Anatomy of the Excel interface (2024)

The main components you will use when exploring and analysing tabular data



- 1 Quick Access Toolbar**
Frequently used commands such as Save, Undo and Redo.
- 2 Title bar**
Shows the workbook name and the Excel application window.
- 3 Search box**
Find commands, help topics and Excel features quickly.

- 4 Ribbon tabs**
Home, Insert, Formulas, Data, Review, View, etc.
- 5 Ribbon**
Context-sensitive command area organised into groups.
- 6 Name Box**
Shows the active cell reference; can also jump to a cell/range.

- 7 Formula Bar**
View, enter and edit values or formulas for the active cell.
- 8 Column headings**
Letters identify columns; click to select an entire column.
- 9 Row headings**
Numbers identify rows; click to select an entire row.

- 10 Active cell**
The currently selected cell; its reference appears in the Name Box.
- 11 Table header & filters**
Field names plus sorting/filter controls in an Excel Table.
- 12 Worksheet grid**
The cell area where tabular data, formulas and results live.

Also important (not visible in this screenshot): sheet tabs, horizontal/vertical scroll bars, the status bar and zoom controls.

ExcelFormulasIn Italian and English

The main matchings

English Excel	Italian Excel	Where we use it
IF	SE	Checks: every flag
AND / OR	E / O	Checks: mortgage vs payment, age, date, score
COUNT	CONTA.NUMERI	Statistics: N numeric values
COUNTA	CONTA.VALORI	Key: total injected anomalies
COUNTBLANK	CONTA.VUOTE	Statistics: blank cells; sentinels and blanks
COUNTIF	CONTA.SE	Statistics: values beyond the fences; Checks: duplicate ID, summary
COUNTIFS	CONTA.PIÙ.SE	Checks: duplicate record; cluster (48 × 199)
AVERAGE	MEDIA	Statistics: mean; z-score
MEDIAN	MEDIANA	Statistics: median
STDEV	DEV.ST	Statistics: standard deviation; z-score
MIN / MAX	MIN / MAX	Statistics; range checks
QUARTILE	QUARTILE	Statistics: Q1, Q3 → IQR
SUMPRODUCT	MATR.SOMMA.PRODOTTO	Checks: N flags and summary
ABS	ASS	Checks: date vs tenure
INT	INT	Checks: date vs tenure

DATE	DATA	Checks: extraction date DATE(2026,9,13)
YEAR	ANNO	Checks: plausible date
TODAY	OGGI	Key: filtering future dates
DATEDIF	DATA.DIFF	Key: date/tenure consistency
LEN	LUNGHEZZA	Key: trailing spaces in categories
TRIM	ANNULLA.SPAZI	Key: trailing spaces in categories
UPPER	MAIUSC	Key: standardising categories

Three modern variants participants may type instead of the "classic" functions used in the file: STDEV.S → DEV.ST.C, QUARTILE.INC → INC.QUARTILE, STANDARDIZE → NORMALIZZA (computes the z-score directly).

The same formula in both languages, separators included — English Excel uses `,` between arguments and `.` for decimals, Italian Excel uses `;` and `,`:

`=IF(OR(B2<18,B2>100),"Implausible age","")` ↔ `=SE(O(B2<18;B2>100);"Implausible age";"`) and `=Q1-1.5*IQR` ↔ `=Q1-1,5*IQR`.

Two main behaviours when editing a formula

Especially regarding the arrow keys

When you are editing a formula in Excel, there are **two main behaviors**, especially for the arrow keys:

- **Edit mode:** the ← → ↑ ↓ keys **move the cursor inside the formula**, without changing the cell references.
- **Point mode:** the arrow keys **select/move the reference to a cell** in the formula.

The F2 key allows you to switch to the formula text editing behavior.

For example, with:

=A1+\$B\$2

if the cursor is on **A1**:

- in **Edit mode**, pressing → moves the cursor between the characters;
- in **Point mode**, pressing → can change the selected reference from **A1** to **B1**.



Edit mode

Arrow keys move the cursor inside the formula.

fx =A1|+\$B\$2



Moves the cursor within the formula text.



Point mode

Arrow keys select/move the reference to a cell.

fx =B1+\$B\$2

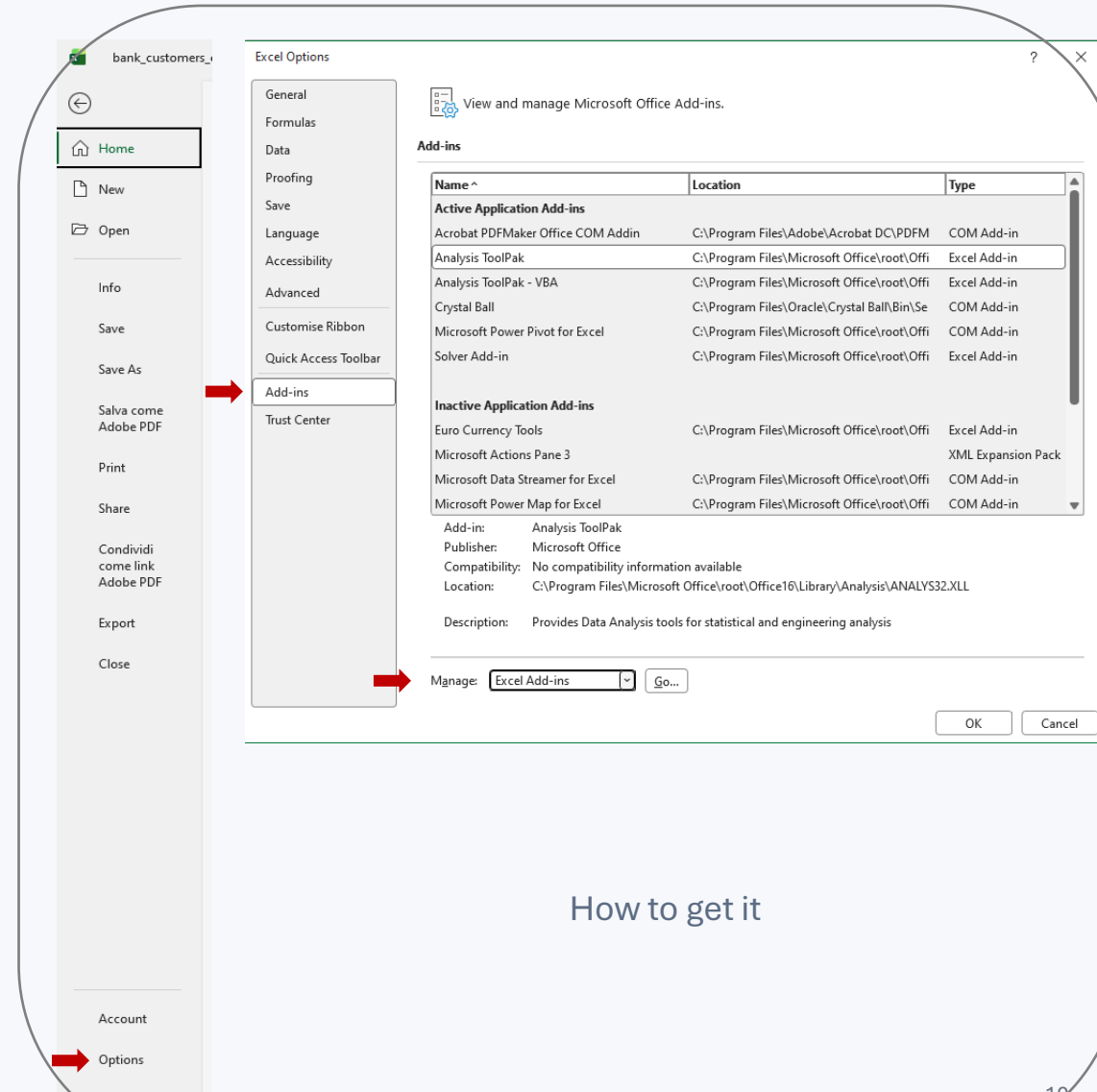
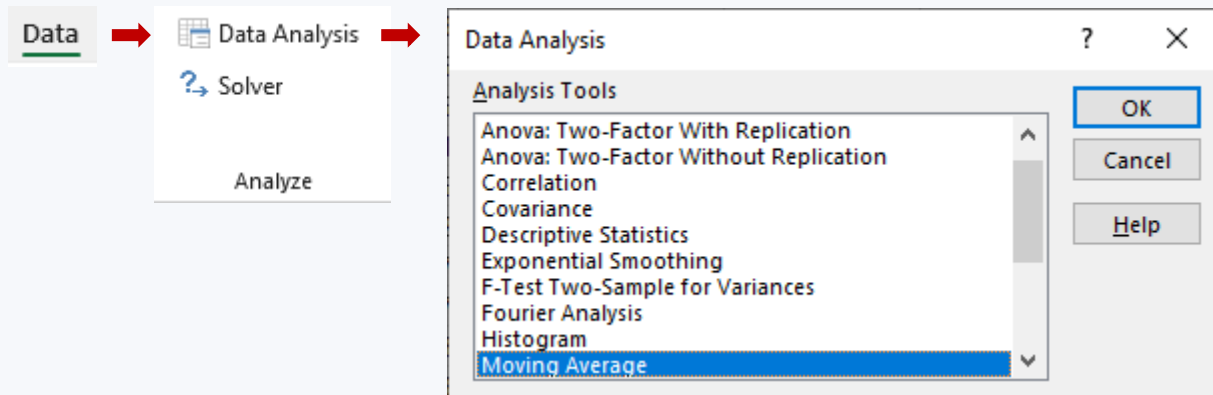
	A	B	C
1	A1	B1	
2			
3			



Changes the selected reference (from A1 to B1).

Analysis tool –a key piece of functionality

What it is



How to get it

Data Science in one slide

Data Science combines business questions, data, statistical reasoning and computational tools to support decisions.



What matters most

A model or dashboard is useful only if it answers a relevant question with trustworthy data.

Where Excel fits

Excel is excellent for inspection, quick EDA, small-to-medium datasets, validation and communication.

Where Excel stops

Very large datasets, heavy automation and advanced modelling often need SQL, Python, R or cloud platforms.

Question → Data → Evidence → Decision

Why Excel still matters for bank analysts

Excel is not “less scientific”; it is a practical analysis surface when used with disciplined methods.

Fast inspection

Open a table and immediately inspect values, types, gaps and extremes.

Business proximity

Analysts, risk teams and finance users already exchange workbooks.

Transparent logic

Formulas, filters and PivotTables make many transformations visible.

Excellent EDA

Descriptive statistics, histograms, scatter plots, box plots and correlations.

Audit support

Checks, reconciliation and exception lists can be made explicit.

Bridge tool

Useful before/after SQL, Python or modelling pipelines.

The key limitation to Data Science is not the tool: it is uncontrolled data handling, hidden assumptions and poor reproducibility.

Data Science, statistics, BI and Machine Learning

They overlap, but they answer different kinds of questions.

Business Intelligence

What happened?

Reports, KPIs, dashboards,
aggregation

Statistics

What can we infer?

Variation, uncertainty, estimation,
testing

Machine Learning

What can we predict?

Prediction, classification,
clustering

Data Science

**How do we turn data into
decisions?**

End-to-end problem framing, data,
analysis, modelling,
communication

Excel can support all four — but EDA and descriptive statistics are where it is especially strong.

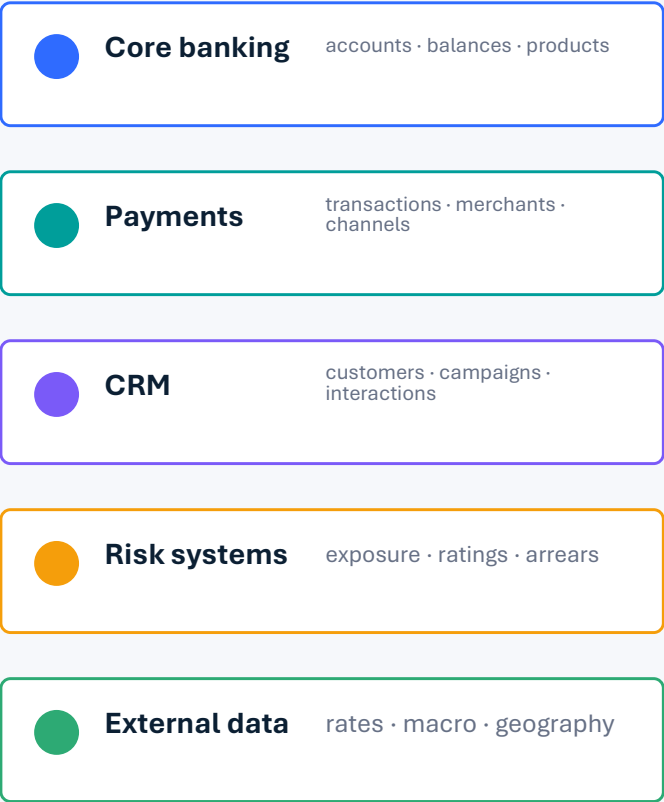
Banking examples: questions before tools

Start with the decision or risk question, then choose the analysis.

Domain	Business question	Excel-first EDA
Retail customers	Which segments show declining balances?	PivotTable + line chart
Credit risk	Which variables move with arrears (ie, payments that are overdue or unpaid when due) or default?	Correlation + distributions
Operations	Where are processing times unusually high?	Histogram + outlier analysis
Fraud monitoring	Which transactions are atypical?	Patterns + anomaly flags
Data governance	Where is information incomplete or inconsistent?	Missingness + quality checks

The data landscape in a bank

Most analysis starts from structured operational data — but meaning comes from context.



x Analysis-ready extract				
fx =ROWS (Table1)				
CustomerID	Date	Balance	Segment	Default
C1001	2026-08-31	12,500	Affluent	No
C1002	2026-08-31	8,200	Mass	No
C1003	2026-08-31	45,500	Affluent	No
C1004	2026-08-31	3,200	Mass	No
C1005	2026-08-31	165,000	Affluent	Yes
C1006	2026-08-31	9,700	Mass	No

A structured table ready to be analysed

Structured vs semi-structured vs unstructured data

Data Science works with many kinds of data. Excel is strongest with structured data, but analysts often receive all three forms.

STRUCTURED DATA

Clear rows, columns and fields

Examples: Excel tables, CSV files, relational databases, banking transaction tables
Typical fields: Customer_ID, Region, Balance, Date, Product_Count
Excel fit: excellent

Recognition: every row represents the same type of entity, and every column has a stable meaning.

SEMI-STRUCTURED DATA

Some organisation, but not fully tabular

Examples: JSON, XML, logs, email headers, API responses
Has labels or tags, but the shape may vary from record to record
Excel fit: possible after transformation or flattening

Recognition: records contain keys or tags, but not every record must contain the same fields.

UNSTRUCTURED DATA

No fixed table structure

Examples: PDFs, emails, contracts, images, audio, call recordings
Meaning exists, but structure is implicit rather than explicit
Excel fit: weak unless data is first extracted or summarised

Recognition: humans can understand the content, but software must work harder to extract variables.

Excel takeaway: structured data is the natural starting point for EDA in Excel; semi-structured and unstructured data usually need preparation before they can be analysed as tables.

Structured data: the backbone of everyday Data Science

Structured data has an explicit schema: rows are observations, columns are variables, cells are values.

x

Table1 · Customer snapshot

fx

=AVERAGE(Table1[Balance])

CustomerID	Age	Income	Balance	Segment
C1001	42	65,000	12,500	Affluent
C1002	35	54,000	8,200	Mass
C1003	61	92,000	45,500	Affluent
C1004	29	—	3,200	Mass
C1005	47	71,000	165,000	Affluent

Rows = observations

One customer, account, loan, transaction or time period per row.

Columns = variables

Age, balance, product, segment, status, date, risk score...

Schema = meaning

Names, types, units, valid ranges, keys and business definitions.

This in Data Science is TIDY DATA

Anatomy of a table

Understanding the grain of the table prevents many analytical mistakes.

x Monthly customer snapshot			
fx =COUNTIFS(Table1[CustomerID],"C1001")			
CustomerID	Month	Balance	Products
C1001	Jan	11,800	3
C1001	Feb	12,100	3
C1001	Mar	12,500	3
C1002	Jan	8,700	2
C1002	Feb	8,450	2
C1002	Mar	8,200	2

Primary key / identifier

CustomerID + Month uniquely identifies each row.

Grain

One row = one customer in one month.

Risk of wrong grain

Averaging balances at the wrong level can double-count customers.

Observation, variable, value: three simple words, three different roles

A table becomes easier to reason about when these terms are explicit.

1

OBSERVATION

A unit being analysed

Example: Customer C1003

2

VARIABLE

A measured attribute

Example: AnnualIncome

3

VALUE

A concrete measurement

Example: €92,000

In Excel: rows usually represent observations, columns variables, and cells values.

What makes tabular data analysis-ready?

A clean table is a model of the business process, not merely a rectangular range.



One row, one observation

No merged cells; no subtotals inside the data area.



One column, one variable

Do not mix currency, text and comments in the same field.



Consistent units

EUR vs thousands of EUR must be explicit.



Stable categories

“Retail”, “RETAIL”, “retail” should not be three categories.



Explicit missing values

Blank ≠ zero ≠ not applicable.



Traceable definitions

Every field has a business meaning and owner.

Tip: convert ranges to an Excel Table (Ctrl+T) before serious EDA.

Why convert a range to an Excel Table?

Before starting EDA, turn a normal cell range into a formal Excel Table (Ctrl+T).

NORMAL RANGE

Example: A1:F1000

ID	Age	Income	Bal.	Seg.
C01	42	65k	12.5k	A
C02	35	54k	8.2k	M
C03	61	92k	45.5k	A
C04	29	—	3.2k	M

Typical formula:
`=AVERAGE(D2:D1000)`



Ctrl+T

EXCEL TABLE

A structured object designed for analysis

- **Stable headers** Column names remain explicit and usable.
- **Built-in filters** Sort and filter every column immediately.
- **Automatic expansion** New rows and formulas are included automatically.
- **Structured references** Use readable formulas such as `Table1[Balance]`.

```
=AVERAGE (Table1[Balance])
```

For EDA, an Excel Table makes the analysis more robust, readable and less error-prone.

How to convert an Excel Table back to a range

The data are preserved; only the Table structure is removed.

1

Select the Table

Click any cell inside the Excel Table.

2

Open Table Design

The Table Design tab appears on the Ribbon.

3

Convert to Range

Choose Table Design → Convert to Range.

4

Confirm

When Excel asks for confirmation, click Yes.

What changes?

Excel Table**Normal range****Data remain**

Values and formulas are preserved.

Formatting

Table formatting usually remains.

Table features

Automatic expansion and structured Table behavior are removed.

Range → Ctrl+T → Excel Table → Table Design > Convert to Range → Range

Data types: meaning comes before formatting

The type determines which operations are meaningful — and which are nonsense.

Numeric

Measured quantities

Balance, age, income, interest rate

Categorical

Labels / groups

Segment, product type, branch

Boolean / binary

Two states

Default yes/no, active/inactive

Date / time

Time coordinates

Transaction date, month-end

Text

Free-form strings

Comments, descriptions, notes

Identifier

Keys, not quantities

CustomerID, IBAN, account number

A CustomerID that contains digits is still an identifier, not a numeric measurement.

Numeric data: discrete and continuous

Both are numeric, but the measurement scale and distribution may differ.

Discrete numeric variables

Countable values, often integers.

Examples: number of products, missed payments, complaints.

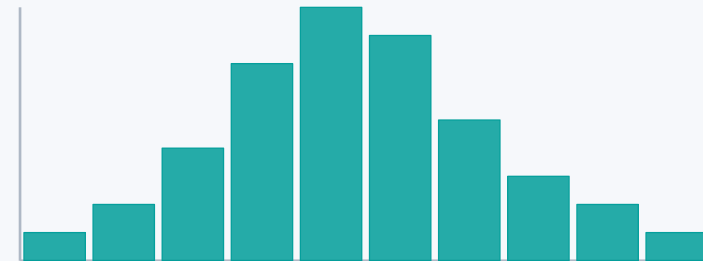


Discrete: Products per customer

Continuous numeric variables

Measurements on a scale.

Examples: balance, income, amount, duration, interest rate.



Continuous: Account balance

```
=AVERAGE (Table1[Balance])
```

```
=MEDIAN (Table1[Balance])
```

```
=STDEV.S (Table1[Balance])
```

Categorical data: nominal, ordinal and binary

Categories can be labels, ordered levels or two-state flags.

Nominal

No intrinsic order

Region: North / South

Product: Card / Mortgage / Loan

Ordinal

Meaningful order

Risk grade: Low / Medium / High

Satisfaction: 1...5

Binary

Exactly two states

Default: Yes / No

Active: 1 / 0

Excel implications

- Use Data Validation lists to control allowed categories.
- Use PivotTables to compare counts, averages and rates across groups.
- Do not average arbitrary codes (e.g., Region 1, 2, 3).

Data types: qualitative and quantitative

A practical taxonomy for structured (tabular) data — with examples and software mappings.

VARIABLES IN TABULAR DATA

QUALITATIVE DATA

aka **categorical**

Values are labels or classes. The set of possible values is often finite and relatively small. Numeric operations such as sums or divisions are not meaningful.

NOMINAL

Unordered categories

No intrinsic ranking between values.

Examples: ZIP code, eye color, region, product type

ORDINAL

Ordered categories

Values have an order, but gaps are not numeric distances.

Examples: rating, risk class, customer tier

QUANTITATIVE DATA

aka **numeric**

Values represent quantities. Arithmetic and distances are meaningful. The theoretical set of possible values may be very large or infinite.

DISCRETE

Counts

Typically integer-valued observations.

Examples: # children, # transactions, # defects

CONTINUOUS

Measurements

Can take any value within an interval, conceptually.

Examples: temperature, height, balance, interest rate

Software mapping in Excel

[See the next slide](#)

Important: A numeric-looking code is not necessarily numeric data. ZIP code and Customer_ID are identifiers.

Key idea: data type is about meaning and valid operations — not only about how a value is formatted in Excel.

Statistical Data Types vs Excel Data Types

Excel stores values using technical cell types. The analyst assigns the statistical meaning.

Statistical data type	Meaning	Typical Excel storage	Example
Nominal categorical	Labels with no inherent order	Text	Region = "Lombardy"
Ordinal categorical	Categories with a meaningful order	Text or coded Number	Low / Medium / High
Discrete numeric	Counts; integer-valued measurements	Number (integer value)	Num_Products = 4
Continuous numeric	Measurements on a continuous scale	Number (decimal value)	Balance = 5,823.47
Binary / Boolean	Two possible states	TRUE/FALSE or 0/1	Defaulted = FALSE
Date / Time	Temporal values	Date/Time*	13/09/2026
Identifier	A code, not a quantity	Usually Text	Customer_ID = "CL00015"
Missing value	Unknown / not recorded	Blank cell	empty cell

* Excel stores dates/times internally as serial numbers, but formats them as dates/times.

KEY IDEA

Excel data type ≠ statistical data type

EXAMPLE 1 — ORDINAL CODES

1 = Low 2 = Medium 3 = High

Excel sees numbers; the analyst sees ordered categories. An average may be mathematically possible but analytically questionable.

EXAMPLE 2 — IDENTIFIERS

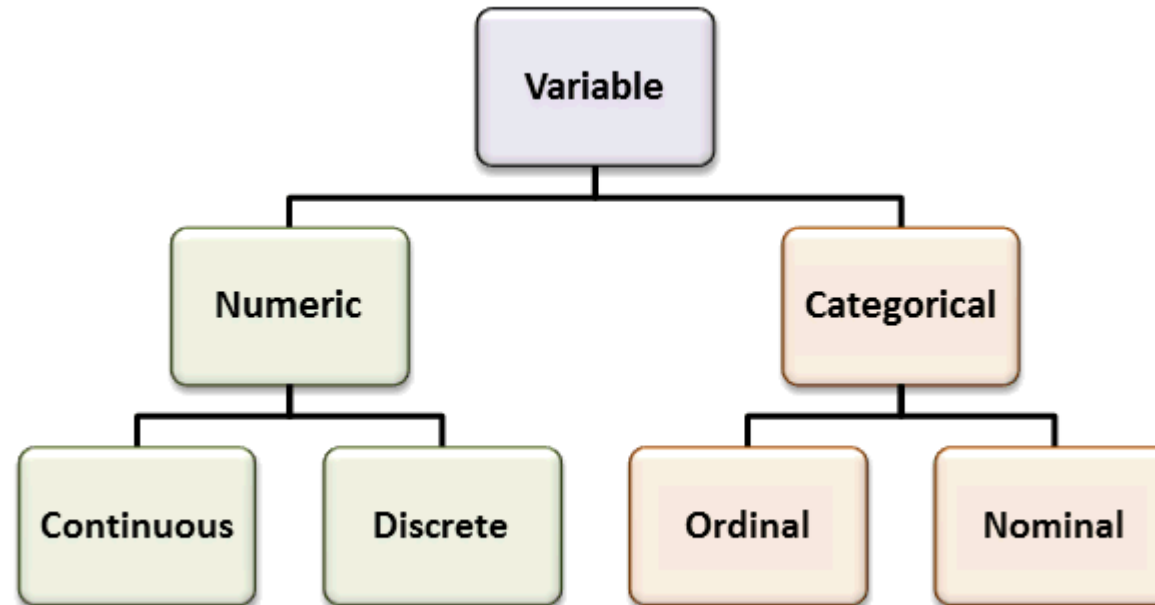
ZIP = 10121 Account = 000173

They look numeric, but they are labels. Store them as Text when leading zeros or code semantics matter.

Analyst rule: determine the variable’s meaning first; only then decide the Excel representation and the valid operations.

Numerical vs Categorical Data Types

Not every column that looks like a number should be treated as a number.



Dates, text and identifiers need special care

Not every column that looks like a number should be treated as a number.

Example	Semantic type	Appropriate analysis
2026-08-31	Date	Sort, group by month/quarter, compute intervals
C100245	Identifier	Join / lookup key — never average it
0012345678	Identifier stored as text	Leading zeros matter
“Mortgage renewal”	Text	Search, classify, count terms
12.5%	Numeric rate	Store as 0.125; display as percentage

Formatting changes appearance; it does not automatically correct the underlying data type.

Common data type problems in Excel

Many “analysis errors” are actually type errors.

x

Type issues hiding in plain sight

fx

=ISTEXT (C4)

CustomerID	Date	Income	Rate	Balance
00125	31/08/26	65,000	4.25%	12,500
00126	08/31/26	54.000	0,0425	8,200
00127	text	92,000	4.10%	—
00128	31/08/26	71k	4.35%	165,000

Symptoms

Sort order looks wrong · SUM ignores values · dates do not group.

Checks

ISTEXT · ISNUMBER · DATEVALUE · VALUE · NUMBERTVALUE.

Prevention

Define schema, use Tables, validation, Power Query types and controlled imports.

=ISNUMBER (C2)
=NUMBERTVALUE (C2)

=ISTEXT (C2)

=DATEVALUE (B2)

Exploratory Data Analysis (EDA): learn the data before modelling it

EDA is a disciplined conversation with the dataset: structure → distributions → relationships → problems → hypotheses.

STRUCTURE

rows · columns · types · grain

DISTRIBUTIONS

centre · spread · shape

RELATIONSHIPS

correlation · groups · trends

DATA ISSUES

missing · outliers ·
inconsistencies

PATTERNS

segments · seasonality ·
anomalies

EDA output is not just charts

It is a set of findings: what the data represents, what looks suspicious, what relationships exist, and what should be investigated next.

EDA = ask → inspect → visualise → quantify → question → repeat

EDA in practice: one graphical example for each of the five lenses

Each lens asks a different question: how the table is organised, how values are distributed, how variables move together, where the problems are, and what broader regularities emerge.

STRUCTURE

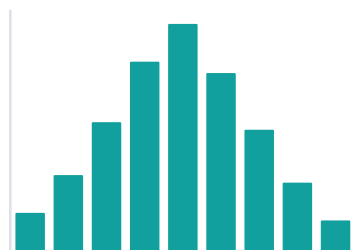
A table with fields and records

ID	Age	Residence	Balance
C0	32	Nor	5,2
C0	41	We	8,1
C0	29	So	3,4
C0	55	Ea	9,7
04		st	80

Example question:
Do we have one row per customer, clear column names and meaningful data types?

DISTRIBUTION

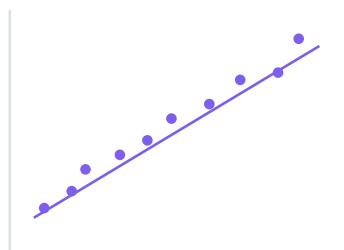
Histogram of a numeric variable



Example question:
How are transaction amounts distributed? Is the shape symmetric, skewed or multi-modal?

RELATIONSHIPS

Scatter plot between two variables



Example question:
Do income and account balance move together? A scatter plot reveals direction and strength.

DATA ISSUES

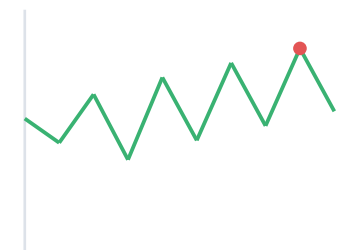
Missing values and outliers

Age	Income	Score
34	28,00	71
41	missi	69
39	31,50	70
36	999,999	72

Example question:
Where are the blanks, errors or extreme values that could distort averages, correlations and charts?

PATTERNS

Trend / seasonality / anomaly



Example question:
Is there a repeated monthly pattern, an upward trend or a point that breaks the usual behaviour?

EDA = structure + distributions + relationships + data issues + patterns

What is a “pattern”?

Pattern recognition means finding recurring structures inside data that may initially look random.

Look at this set of numbers and letters:

- knsxoi543y77778efh8rfv097777d
- j39e8389f7777lmjbh8ihrz7777
- knsxoi543y77778efh8rfv097777
- 39e8389f7777lmjbh8ihrz>7777

Do you notice anything unusual?

Exactly. After every 10 apparently random letters and numbers, the digit “7” is repeated four times — the pattern “7777”. Your brain has recognised a regular structure inside data that initially looks **random**. This process is called **pattern recognition**.

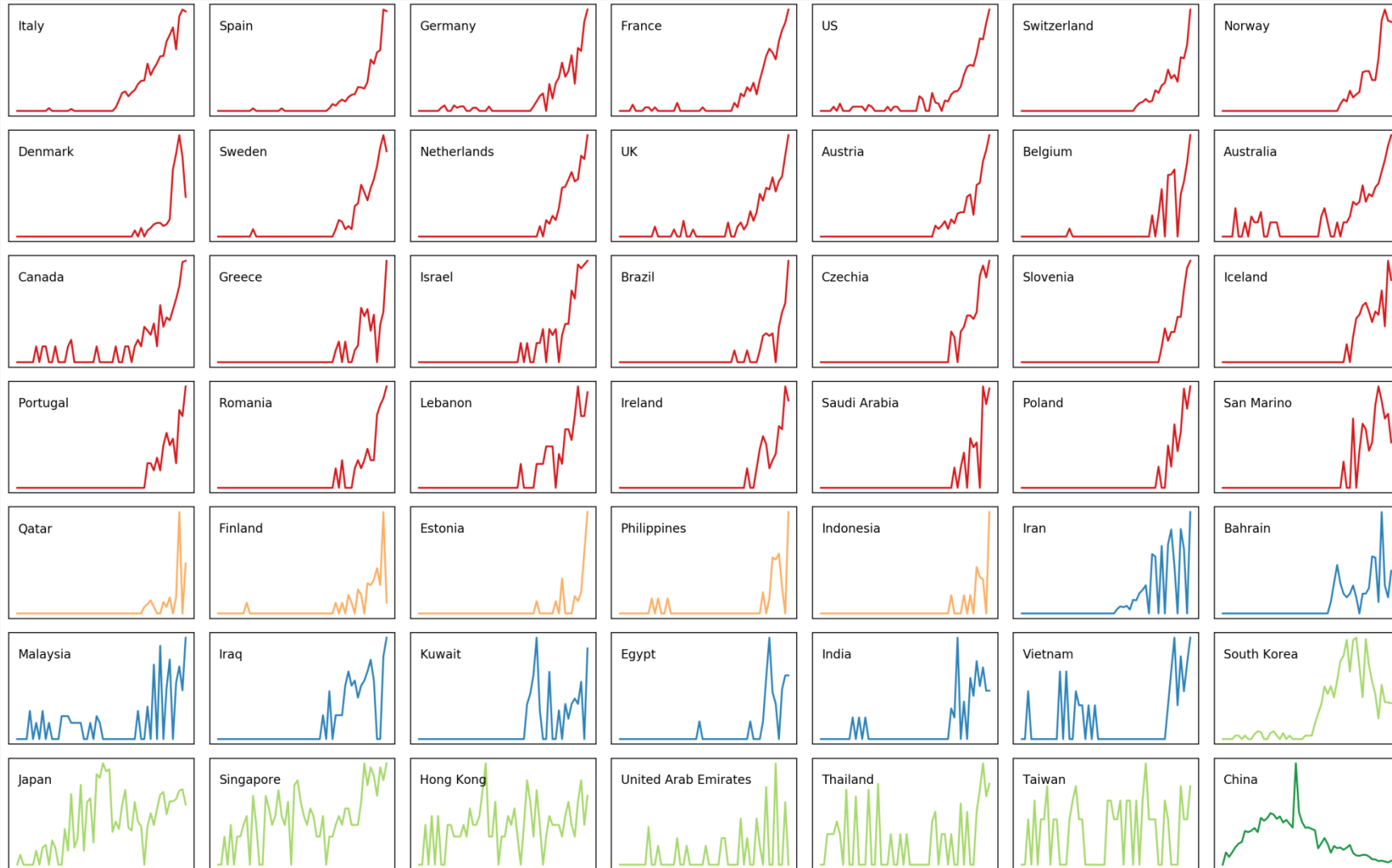
Pattern recognition = **identifying recurring models, structures or regularities within raw data.**

Patterns are repeated structures



The same idea applies to data: repeated forms, relationships, sequences or behaviours can be discovered and described.

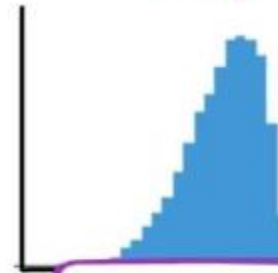
Patterns



Patterns in Covid-19 growth

EDA

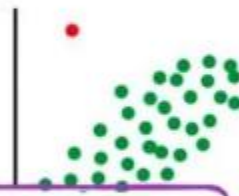
Exploratory Data Analysis



Data distribution

	F	G	H	I	J
A	0.620576	0.140053	1.352728	NaN	0.808078
B	NaN	0.526829	NaN	NaN	0.170902
C	NaN	0.458827	1.406713	0.071119	NaN
D	NaN	2.307197	NaN	NaN	NaN
E	0.203402	0.259913	NaN	0.505811	1.516755

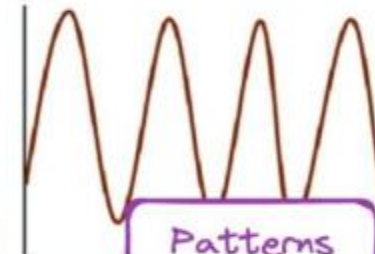
Missing data



Outliers



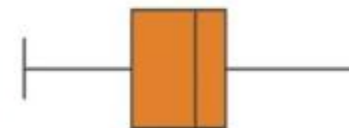
Correlation



Patterns

```
Cust_No      int64
Cust_Name    object
Product_id   int64
Product_cost float64
Purchase_Date datetime64[ns]
dtype: object
```

Data types



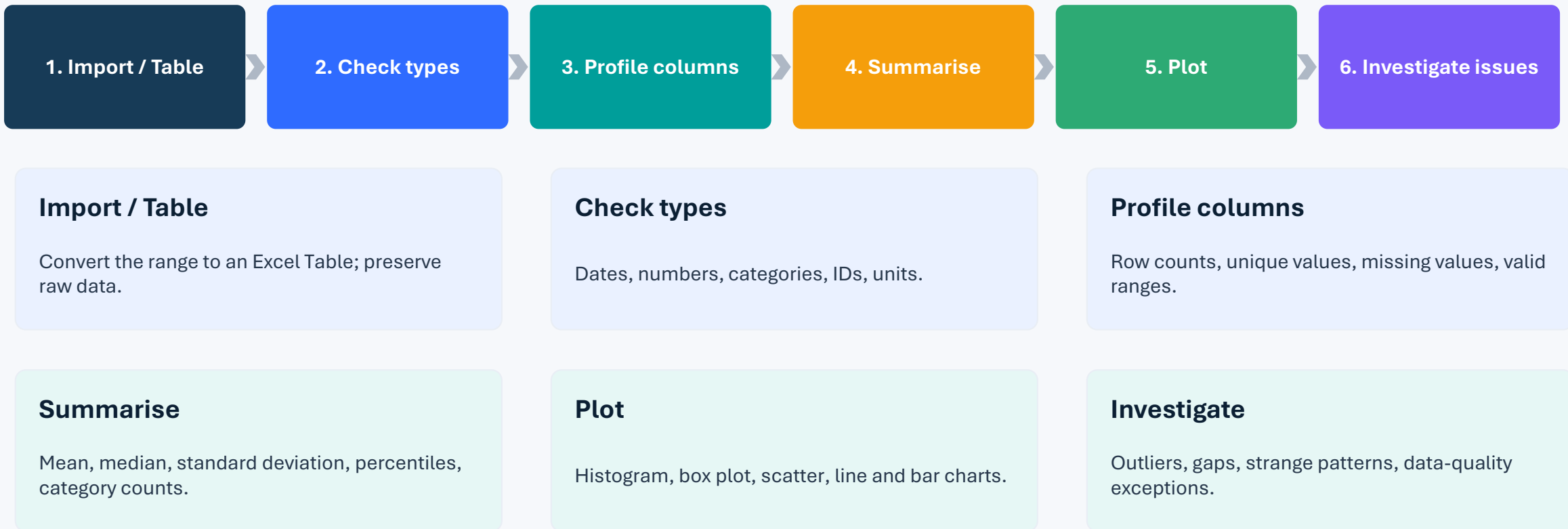
Data visualization



Data quality

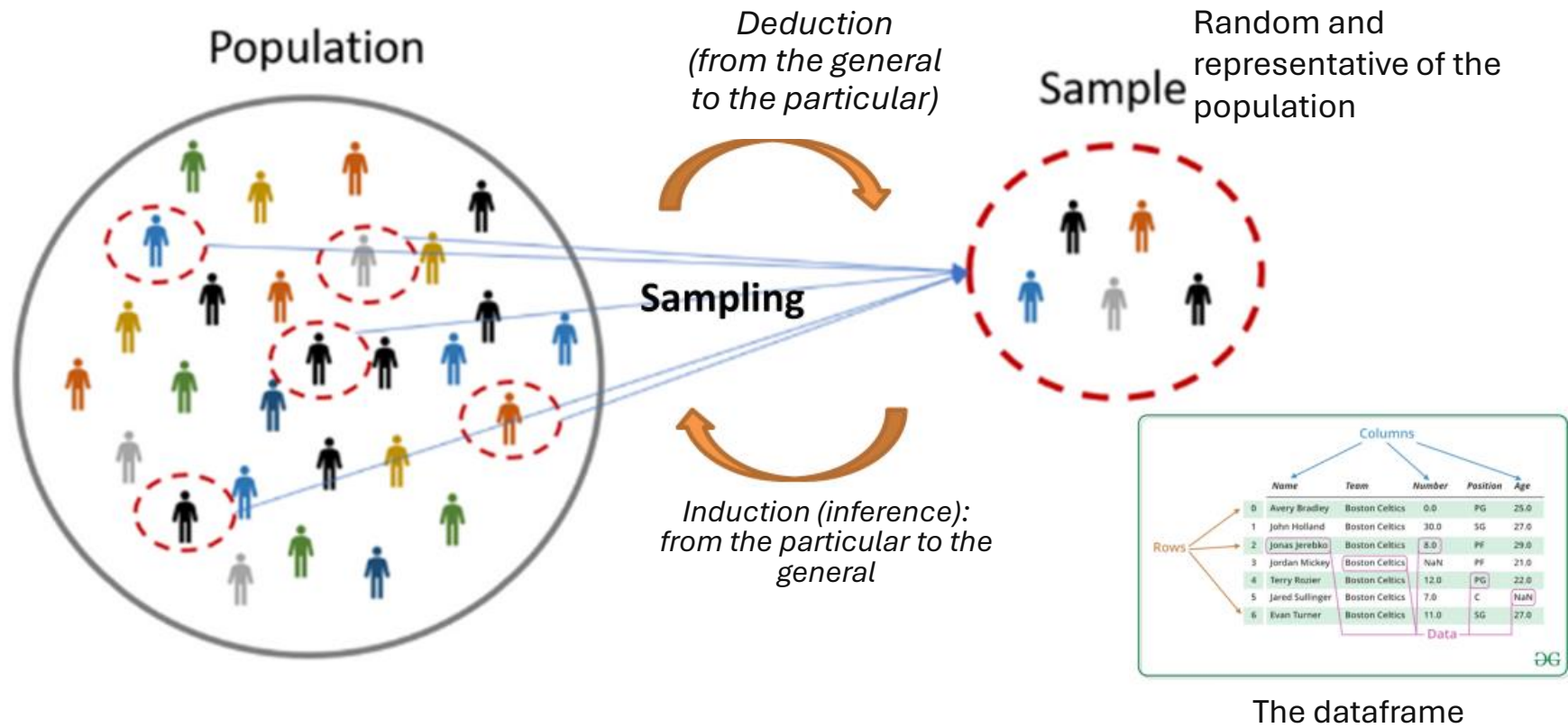
A practical EDA workflow in Excel

Use a repeatable sequence so important checks are not skipped.



Best practice: keep “raw”, “clean”, “analysis” and “output” areas separate.

Sample vs. Population



Sampling selects observations from the population; statistical inference uses the sample to learn about the population.

Building a dataset in Excel: two starting points

A dataset can be created directly in Excel or imported from an external source. In both cases, the goal is a clean, structured, analysis-ready table.

1 BUILD THE DATASET NATIVELY IN EXCEL

Typical sources

- Manual data entry
- Copy / paste from controlled sources
- Form-based or operational data entry

Customer_ID	Age	Region	Balance
CL001	34	North	5,240
CL002	41	West	8,120
CL003	29	South	3,410

Good practice

- One row = one observation
- One column = one variable
- Use clear headers and consistent data types
- Convert the range to an Excel Table (Ctrl+T)

OR

2 IMPORT THE DATASET FROM OUTSIDE EXCEL

Typical external sources

- CSV / TXT files
- Other Excel workbooks
- Databases
- Web / API sources via Power Query

CSV / DB

Data →
Get Data

Review import
settings

Check before loading

- Delimiter and text encoding
- Scale (decimal and date conventions)
- Truncated data types
- Missing values and unexpected columns

ANALYSIS-READY
EXCEL TABLE

Key point: regardless of the starting point, verify structure, data types, missing values and business meaning before starting EDA.

CSV files: simple tabular data in plain text

CSV stands for Comma-Separated Values: a lightweight format for storing and exchanging structured data.

customers.csv

```
Customer_ID, Age, Income, Balance, Default  
CL0001, 42, 65000, 12500, No  
CL0002, 35, 54000, 8200, No  
CL0003, 61, 92000, 45500, No  
CL0004, 29, 45000, 3200, Yes
```

Each line is a row. Separators divide fields into columns. The first row usually contains column names.

Plain text

No formulas, charts, formatting or multiple worksheets — only values.

Tabular structure

Rows represent observations; fields represent variables.

Delimiter matters

Despite the name, CSV files may use commas, semicolons or other delimiters depending on locale and source.

CSV is a data format, not a spreadsheet workbook.

Why CSV is useful in Data Science

CSV is widely used as a simple interchange format between analytical tools, databases and programming environments.

Portable

Readable by Excel, Python, R, SQL tools, BI platforms and many other applications.

Transparent

The content can be inspected with a text editor: there is little hidden structure.

Easy to exchange

Useful for exporting datasets from systems and moving tabular data between tools.

Automation-friendly

Straightforward to read and write in scripts and data pipelines.

Lightweight

No workbook features are stored, so the format focuses on the data itself.

Good for datasets

A natural representation for rows × columns used in EDA and Machine Learning.

Typical Data Science flow: source system → CSV → Excel / Python / R → analysis

CSV

CSV ↔ Excel: how to convert in both directions

Import a CSV when you want to analyse it in Excel; export to CSV when you need a simple, portable data file.

CSV → Excel

1. In Excel, choose Data → From Text/CSV.
2. Select the .csv file.
3. Check delimiter, encoding and the previewed columns.
4. Choose Load to bring the data into Excel.
5. Save as .xlsx if you want to preserve Excel features.

Excel → CSV

1. Keep the worksheet you want to export active.
2. Choose File → Save As.
3. Select CSV UTF-8 (Comma delimited) (*.csv) when appropriate.
4. Save the file and confirm Excel's warnings.
5. Re-open or inspect the CSV if delimiter/encoding is critical.

Important: CSV stores one flat table only — formulas, formatting, charts, PivotTables and additional worksheets are not preserved.

CSV files across locales: when commas stop being simple

A CSV file is plain text. The delimiter, decimal symbol, dates and encoding may be interpreted differently by Excel depending on regional settings.

KEY IDEA Never assume that “CSV” means the same parsing rules on every computer. Import settings must be explicit.

1 What can change with locale?

- **Field delimiter** Comma (,), semicolon (;), tab, etc.
- **Decimal symbol** 3.75 in EN/US vs 3,75 in many EU locales
- **Thousands separator** 25,900.50 vs 25.900,50
- **Date interpretation** 03/04/2026 can mean Mar 4 or Apr 3
- **Text encoding** UTF-8 avoids many accented-character problems

2 Example: the same data, different conventions

Source CSV created in an English/US environment

```
Customer_ID,Income,Rate,Region
C001,25900.50,3.75,Turin
C002,17200.00,4.10,Milan
```



Double-click / automatic opening

POSSIBLE RESULT

- all fields in one column
- decimal values split incorrectly
- numbers stored as text
- dates silently reinterpreted

Explicit import settings

SAFE RESULT

- delimiter set explicitly
- decimal locale set correctly
- columns typed as expected
- row/column counts validated

3 Safe Excel import workflow

1

Data → From Text/CSV

Do not rely on double-click when locale is uncertain.

2

Set delimiter explicitly

Choose comma, semicolon, tab, etc.

3

Check File Origin / UTF-8

Preserve accented characters and symbols.

4

Set locale for numbers & dates

Especially decimal comma/dot and date order.

5

Validate before analysis

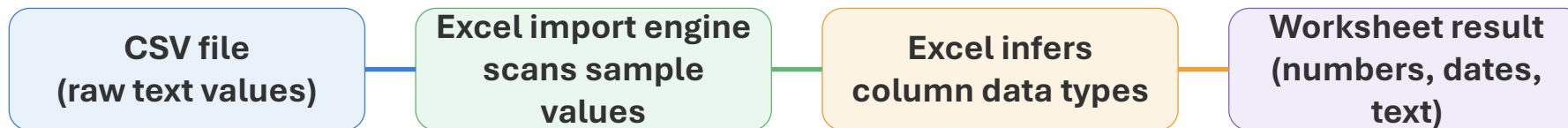
Check column count, data types, totals and sample rows.

For shared workflows: document the delimiter, encoding and locale.

CSV import in Excel: data type inference

Why the same CSV file can be interpreted differently across tools, locales and workflows

WHAT EXCEL DOES



Typical inference examples

CSV values	Inferred as	Possible issue
"00123"	Number	Leading zeros lost
"13/09/2026"	Date	Locale ambiguity
"1,234"	Number/Text	Decimal vs thousands
"TRUE"	Logical/Text	Depends on parser
"NA" / blank	Missing/Text	May not mean the same thing
"CL00015"	Text	Usually OK

Key idea: a CSV file stores text separated by delimiters; data types are reconstructed during import.

Why this matters

The same column may be imported as text, number or date. Wrong inference can distort sorting, filtering, formulas and charts. IDs, ZIP codes and account numbers can be damaged by numeric conversion. Cross-language imports may change decimal separators and date interpretation.

Safer Excel workflow

Use Data → From Text/CSV instead of double-clicking the file.
Review delimiter, file origin / encoding, locale, and detected data type.
Force Text for identifiers; verify dates, decimals and missing-value markers.

Takeaway: CSV files do not carry rich type metadata. Excel infers types during import, so analysts must review the import settings before trusting the data.

First look: create a dataset profile

Before any deep analysis, answer a small set of basic questions.

x

Column profile

fx

=COUNTBLANK (Table1[Income])

Field	Type	Missing	Unique	Min	Max
Age	Numeric	0	41	18	86
Income	Numeric	37	1,204	12k	420k
Segment	Category	0	4	—	—
Balance	Numeric	4	8,951	-1.2k	650k
Default	Binary	0	2	0	1

How many rows?

ROWS, COUNTA or Table totals.

How many unique values?

UNIQUE + ROWS (modern Excel) or PivotTable.

What is missing?

COUNTBLANK, ISBLANK, filters and conditional formatting.

=ROWS (Table1)

=COUNTA (Table1[CustomerID])

=COUNTBLANK (Table1[Income])

Descriptive statistics: centre, spread and shape

Summary statistics compress a distribution into a few interpretable numbers.

Mean

Average level

`=AVERAGE ( range )`

Median

Middle value

`=MEDIAN ( range )`

Std. deviation

Typical spread

`=STDEV . S ( range )`

Min / Max

Observed range

`=MIN ( range ) / =MAX ( range )`

Quartiles

25th / 50th / 75th percentiles

`=QUARTILE . INC ( range , k )`

Important

Mean and standard deviation can be heavily affected by outliers. Always compare with median and quartiles.

The “Descriptive Statistics” option in Excel Analysis ToolPak can generate a compact summary table.

PivotTables: fast multivariate EDA without writing formulas

PivotTables help answer “how does a metric differ across groups?”

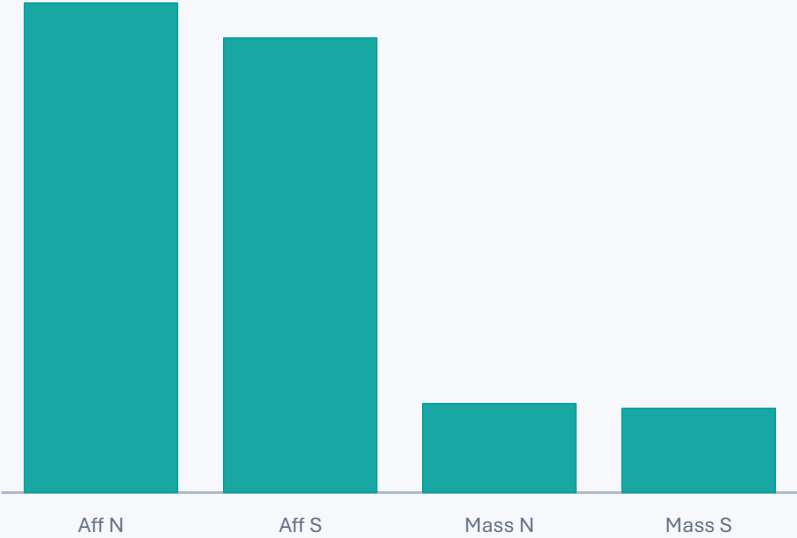
x

PivotTable output

fx

Average of Balance / Count of CustomerID

Segment	Region	Customers	Avg Balance	Default %
Affluent	North	1,240	46,200	0.8%
Affluent	South	980	42,900	1.1%
Mass	North	4,650	8,400	2.9%
Mass	South	4,210	7,950	3.4%



Average balance by segment & region

PivotTables are a bridge between raw tables and clear business questions.

Sorting, filtering and conditional formatting are EDA tools

Simple spreadsheet operations often reveal data issues faster than complex methods.

x Filter & inspect			
fx =SORT (Table1[Balance], , -1)			
Customer	Income	Balance	DaysPastDue
C1001	65,000	12,500	0
C1002	54,000	8,200	0
C1003	92,000	45,500	12
C1004		3,200	0
C1005	71,000	165,000	97
C1006	58,000	9,700	0

Sort

Find top/bottom values and suspicious extremes.

Filter

Inspect one segment, branch, category or exception at a time.

Conditional formatting

Highlight blanks, duplicates, thresholds and unusual values.

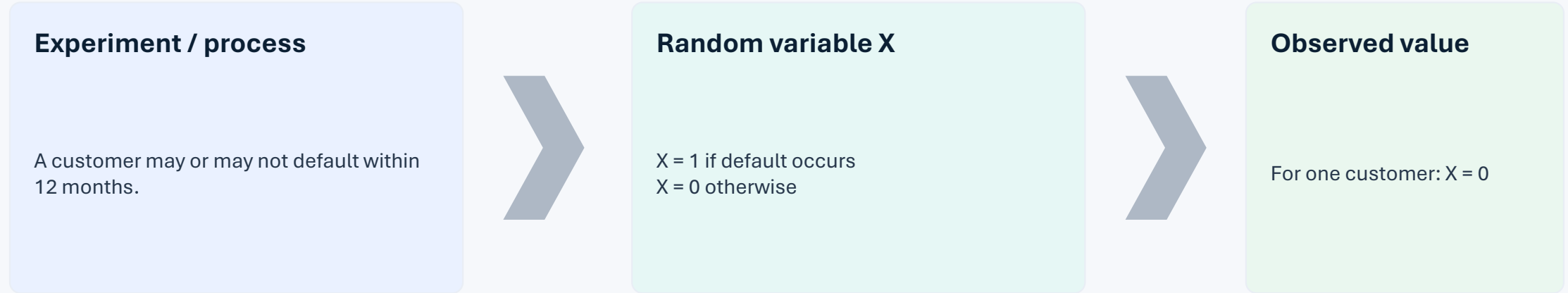
Freeze panes / Table headers

Keep context while scanning many rows.

Tip: “show me the 20 largest values” is often a very good EDA question.

Random variables: a bridge from data to probability

A random variable represents a quantity whose observed value depends on an uncertain outcome.



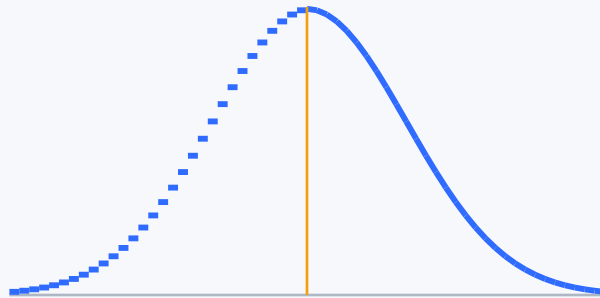
Other examples

Transaction amount · number of late payments · account balance next month · time until loan closure · daily loss.

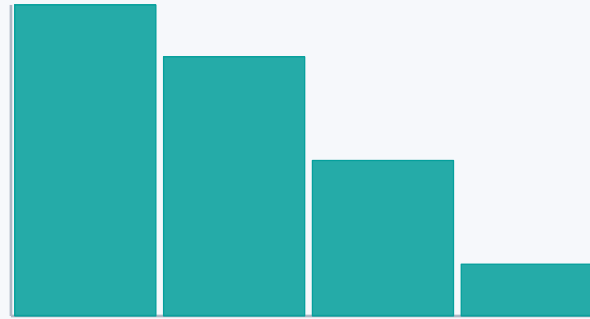
Random does not mean “without structure”. It means the exact outcome is uncertain before observation.

Probability distributions: the “shape” of uncertainty

A distribution describes which values a random variable can take and how likely they are.



Continuous distribution



Discrete distribution

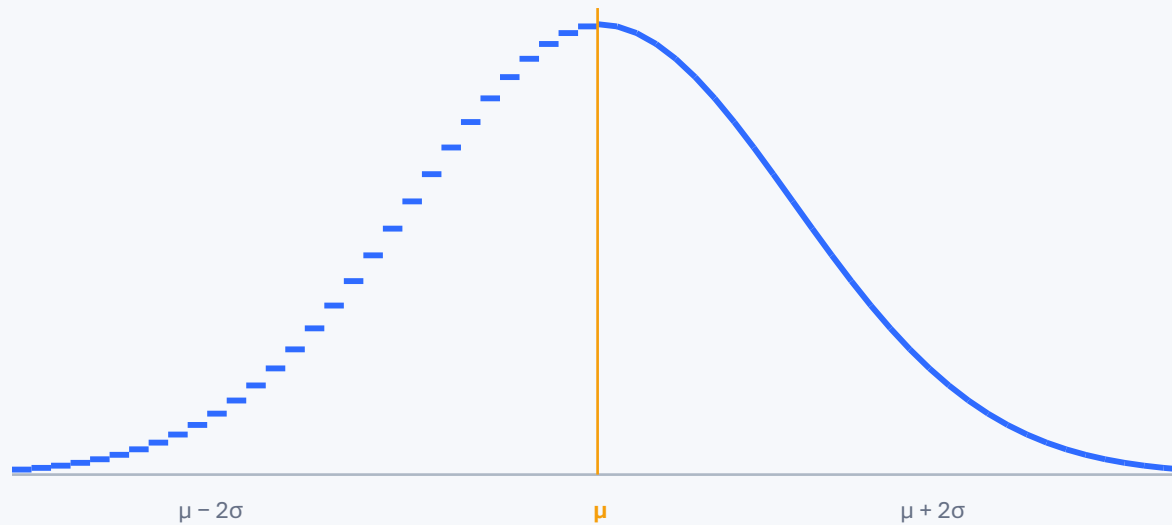


Binary distribution

- Distributions help us reason about typical values, variability and tail risk.
- The empirical distribution comes from observed data; a theoretical distribution is a model.
- EDA asks whether the observed shape is symmetric, skewed, multi-modal, heavy-tailed or irregular.

The Normal distribution: useful model, not universal law

Many measurement errors and aggregate quantities are approximately bell-shaped — but financial data often has skew and heavy tails.



Parameters

Mean μ controls location; standard deviation σ controls spread.

EDA question

Does the histogram look roughly symmetric and bell-shaped?

Banking caution

Losses, balances and transaction amounts are often skewed and may have heavy tails.

Excel examples: `=NORM.DIST(x, mean, stdev, FALSE)` | `=NORM.INV(p, mean, stdev)`

Bernoulli and Binomial: natural distributions for “event / no event”

Binary outcomes appear everywhere in banking: default, fraud flag, churn, approval, response.

Bernoulli random variable

One trial with two outcomes.

$X = 1$ with probability p

$X = 0$ with probability $1 - p$



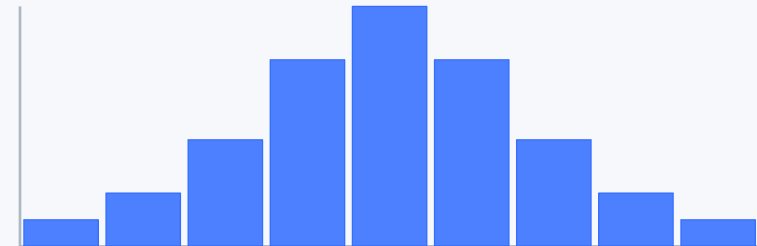
Binomial random variable

Counts how many “successes” occur in n comparable Bernoulli trials.

Example: number of defaults among 100 similar loans.



Bernoulli: one customer



Binomial: count across many customers

Probability vs statistics: opposite directions of reasoning

Probability starts from a model; statistics starts from data.

Probability

MODEL / assumptions



Possible data / outcomes

Statistics

OBSERVED DATA



Model / parameters / inference

EDA sits mainly on the statistics side: use observed data to learn what is plausible, typical and suspicious.

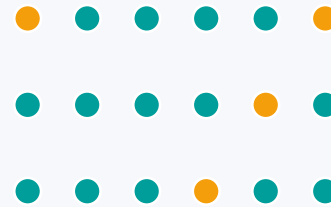
Sampling and uncertainty: a sample is not the population

Any estimate from a sample varies from sample to sample. Statistical thinking quantifies that uncertainty.

Population



Sample



Estimate

Sample mean
Sample default rate
Sample correlation
... each has uncertainty.

- Larger, representative samples usually reduce uncertainty.
- Biased samples can be precise but wrong.
- Always ask: “how was this data selected?”

Correlation: how two numeric variables move together

Correlation quantifies the strength and direction of a linear relationship.

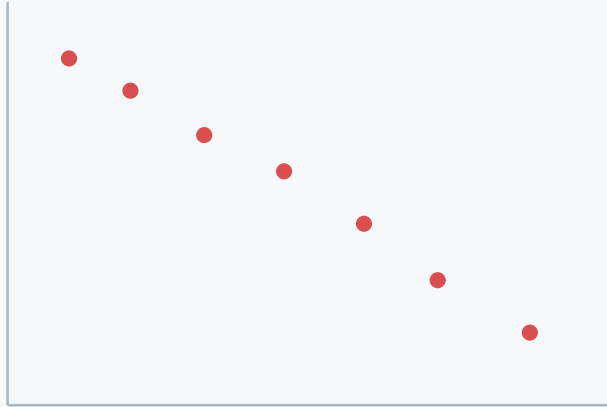
Positive correlation



$$r > 0$$

High values of X tend to occur with high values of Y.

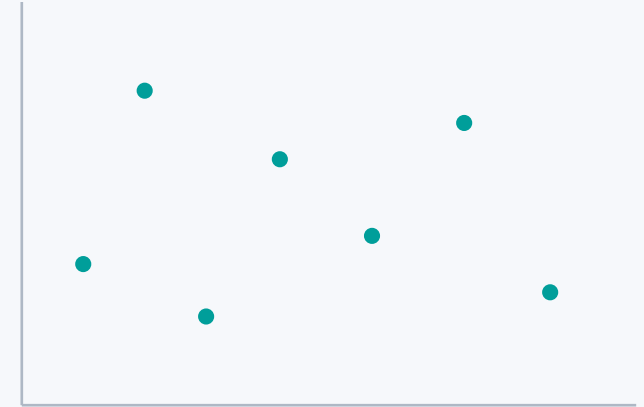
Negative correlation



$$r < 0$$

High values of X tend to occur with low values of Y.

Little linear correlation



$$r \approx 0$$

No strong linear relationship — but nonlinear structure may still exist.

The Pearson correlation coefficient

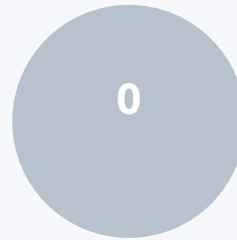
Pearson r is bounded between -1 and $+1$ and measures linear association.



Perfect negative



Moderate negative



No linear relationship



Moderate positive



Perfect positive

Scale-free

Changing EUR to thousands of EUR does not change r .

Linear

r can miss curves, thresholds and other nonlinear patterns.

Sensitive to outliers

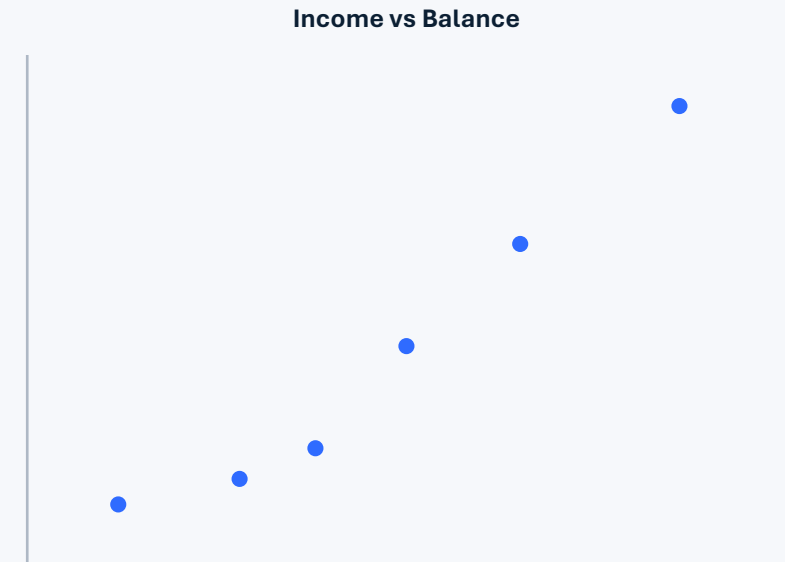
One extreme point can dramatically change the result.

Interpret the coefficient together with a scatter plot.

Correlation in Excel: CORREL + scatter plot

A number is useful; a number plus a plot is much safer.

x Income vs Balance		
fx =CORREL(Table1[Income],Table1[Balance])		
Customer	Income	Balance
C1001	65,000	12,500
C1002	54,000	8,200
C1003	92,000	45,500
C1004	45,000	3,200
C1005	71,000	40,000
C1006	58,000	9,700



=CORREL(Table1[Income], Table1[Balance])

Always check units, missing rows and outliers before interpreting r.

Correlation matrix: many pairwise relationships at once

A correlation matrix is a compact map of linear associations among numeric variables.

	Age	Income	Balance	Products	Arrears
Age	1.00	0.28	0.21	0.15	0.05
Income	0.28	1.00	0.63	0.36	-0.18
Balance	0.21	0.63	1.00	0.42	-0.31
Products	0.15	0.36	0.42	1.00	-0.12
Arrears	0.05	-0.18	-0.31	-0.12	1.00

What to look for

Large absolute values $|r|$ reveal strong linear association.

Diagonal

A variable is perfectly correlated with itself: $r = 1$.

Symmetry

$\text{corr}(X,Y) = \text{corr}(Y,X)$, so the matrix mirrors across the diagonal.

Excel: Data Analysis → Correlation or multiple `=CORREL(...)` formulas

Correlation is not causation

A strong relationship can be real and useful without proving that X causes Y.

Common cause

A third variable drives both X and Y.

Example: season affects both heating usage and energy bills.

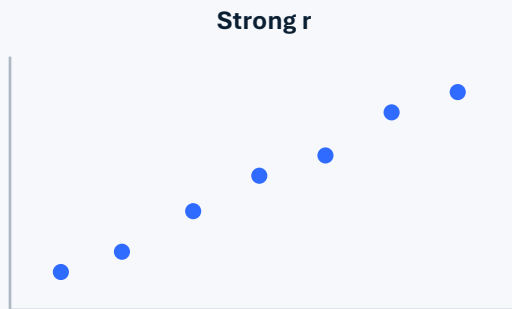
Reverse direction

Y may influence X rather than X influencing Y.

Correlation alone cannot determine direction.

Spurious relationship

With many variables, some correlations appear by chance.



Does X cause Y?

Not answered by correlation alone

Need additional evidence

Domain knowledge · time ordering · experiments · causal designs.

Choose the plot that matches the question

Visualisation is part of reasoning, not decoration.

Distribution of one numeric variable



Histogram / Box plot

Relationship between two numeric variables



Scatter plot

Metric over time



Line chart

Compare categories



Bar / Column chart

Missingness / quality exceptions

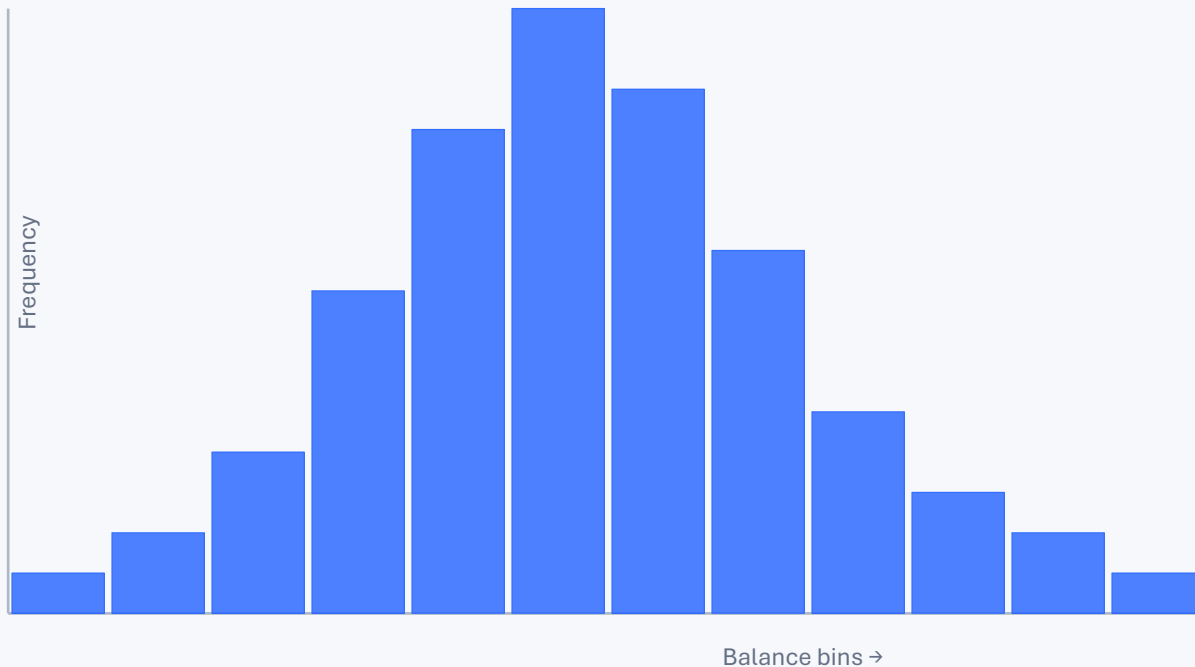


Conditional formatting / table

Avoid charts that hide the data structure. Pie charts, 3D effects and decorative clutter are often poor analytical choices.

Histogram: see the distribution, not only the average

A histogram groups numeric observations into bins and shows how frequently values occur.



Reveals

Skewness · multiple modes · gaps · tails · unusual peaks.

Excel

Insert → Statistical Chart → Histogram, or Data Analysis → Histogram.

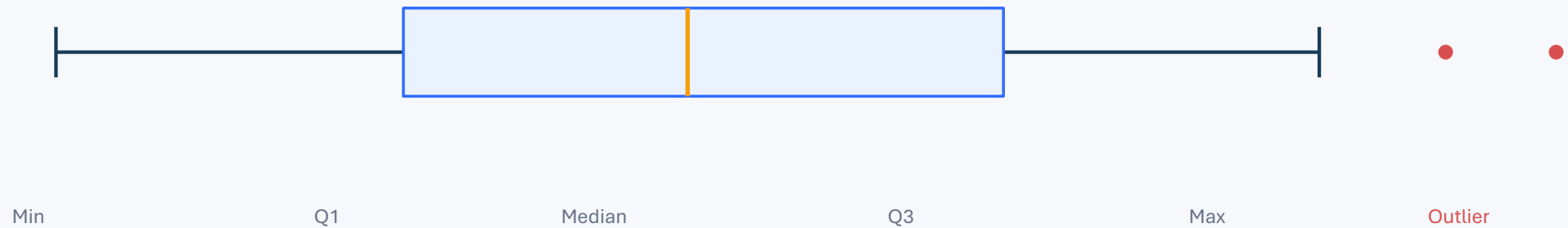
Watch the bins

Too few bins hide structure; too many bins create noise.

Question: “What does the distribution of customer balances look like?”

Box plot: compact view of centre, spread and outliers

The box spans Q1 to Q3, the line is the median, whiskers show the non-outlier range, and points may mark outliers.



Strength

Very fast comparison across customer segments, products or branches.

Robust centre

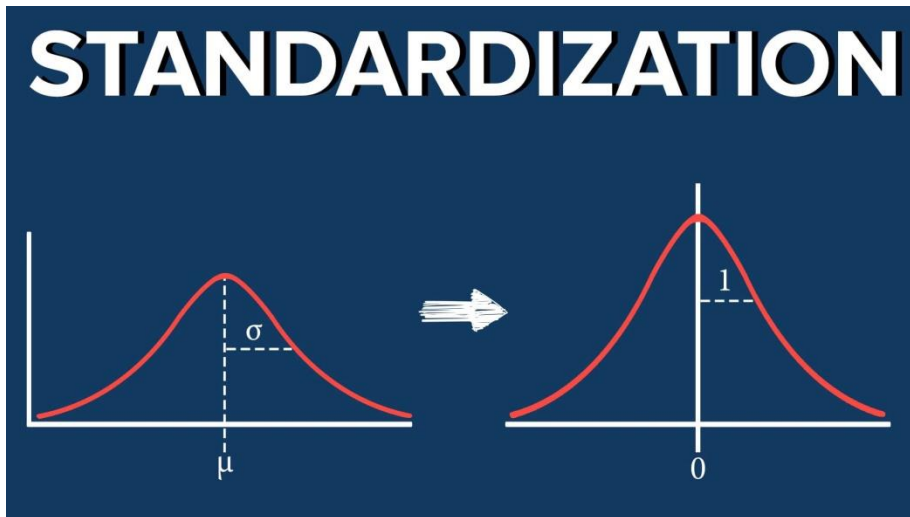
Median and quartiles are less sensitive to extreme values than the mean.

Excel

Insert → Statistical Chart → Box & Whisker.

Box plots: required to compare unevenly distributed data

Two typical useful techniques to compare unevenly distributed data by boxplots → this way, we can then appreciate differences



Normalization Formula

$$X_{\text{normalized}} = \frac{(X - X_{\text{minimum}})}{(X_{\text{maximum}} - X_{\text{minimum}})}$$



Standardising a numeric column in Excel (Z-score)

Standardisation expresses each value as a number of standard deviations from the mean.

CONCEPT

Z-score formula

$$z = (x - \mu) / \sigma$$

Interpretation

z = 0 → value is at the mean
z = +1 → one standard deviation above
z = -1 → one standard deviation below

EXCEL FORMULA

For values in A2:A101

```
= (A2 - AVERAGE ($A$2 : $A$101) )  
/STDEV.S ($A$2 : $A$101)
```

Copy the formula down the column.

EXAMPLE

Annual income (k€)

Original	Z-score
20	-1.26
40	-0.63
60	0.00
80	+0.63
100	+1.26

When standardisation is useful

- Variables have very different units or scales
- Distance-based methods are used (e.g. k-NN, k-means)
- You want a common scale centred around zero

Important

Standardisation does not force values into a fixed range. Values below -3 or above +3 are possible and may highlight unusual observations.

Takeaway: standardisation changes the scale, not the relative ordering of the observations.

Normalising a numeric column in Excel (Min–Max)

Min–Max normalisation rescales a variable to a fixed interval, most commonly 0 to 1.

CONCEPT

Min–Max formula

$$(x - \text{min}) / (\text{max} - \text{min})$$

Result

minimum → 0
maximum → 1
all other values → proportionally between 0 and 1

EXCEL FORMULA

For values in A2:A101

```
= (A2-MIN($A$2:$A$101))  
/ (MAX($A$2:$A$101)-  
MIN($A$2:$A$101))
```

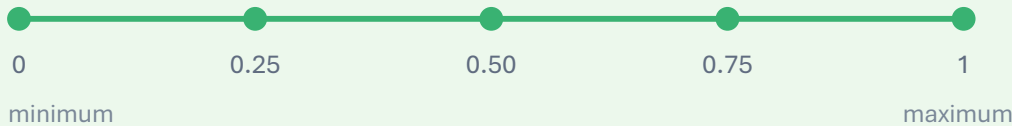
For a 0–100 scale, multiply the result by 100.

EXAMPLE

Customer score

Original	0–1 scaled
20	0.00
40	0.25
60	0.50
80	0.75
100	1.00

How to read the new scale



Watch out for outliers

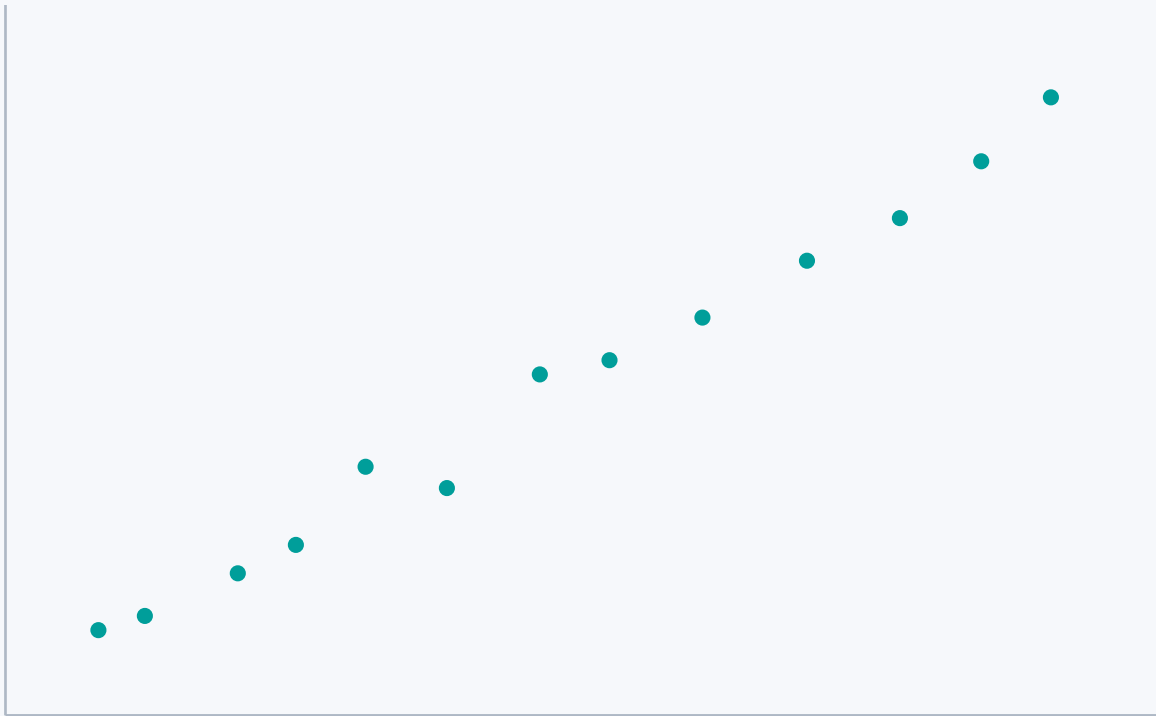
Min–Max normalisation is highly sensitive to extreme minimum or maximum values. One outlier can compress most observations into a very narrow part of the 0–1 interval.

Takeaway: normalisation maps values to a chosen range; standardisation centres and scales them around the mean.

Scatter plot: the essential chart for relationships

Every point is one observation positioned by two numeric variables.

Income vs Balance



Look for

Direction · strength · curvature · groups · isolated points.

Add meaning

Colour points by segment or branch; use labels only for key exceptions.

Excel

Insert → Scatter. Add a trendline only when it supports the question.

Pair with `=CORREL(...)` – never interpret the coefficient without seeing the points.

Line chart: patterns across time

Use line charts when order matters — especially for time series.



Best for

Trend · seasonality · regime shifts · before/after comparisons.

Banking example

Monthly deposits, delinquency rate, call volume or transaction counts.

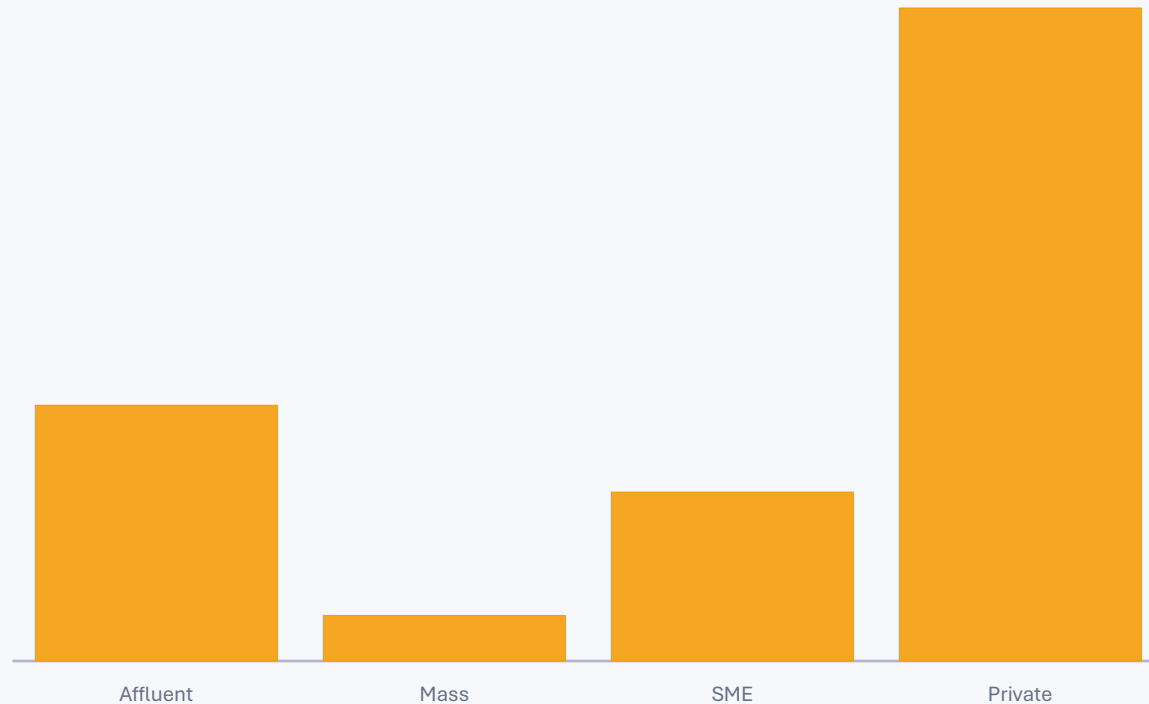
Caution

Do not connect unordered categories with a line.

Excel: put a real date in the X-axis whenever possible.

Bar and column charts: compare categories clearly

The baseline should normally start at zero because bar length encodes magnitude.



Use for

Counts, averages, totals and rates by category.

Good practice

Sort bars when ranking matters; keep category labels readable.

Avoid

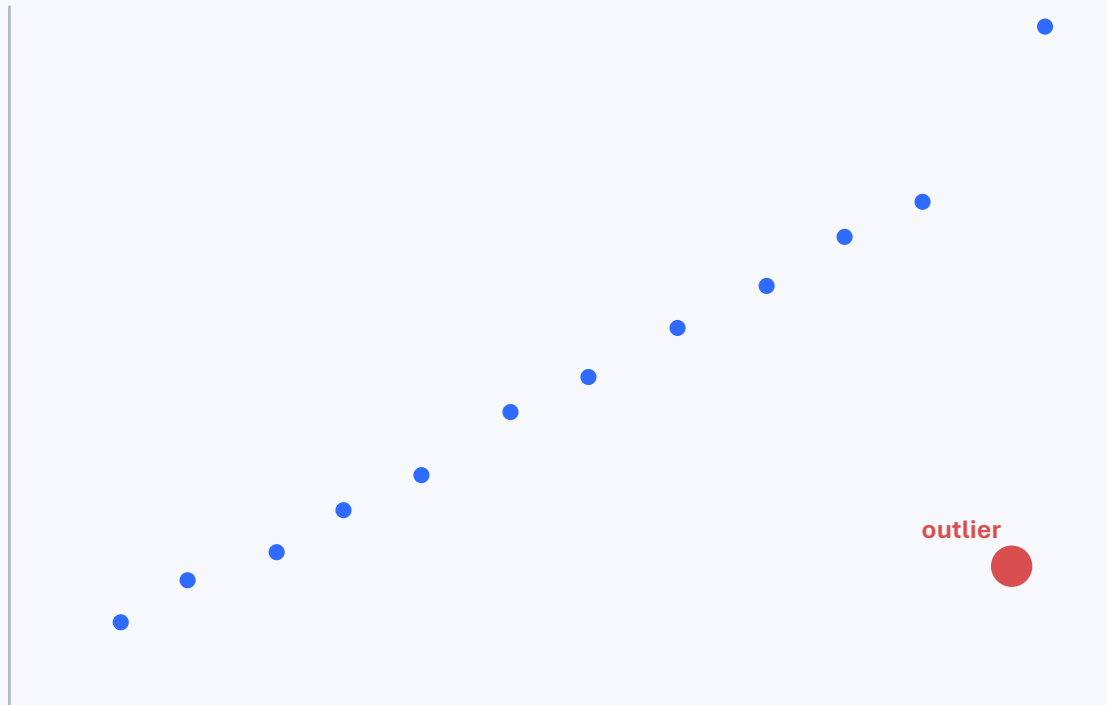
3D bars, too many categories and truncated baselines.

Question: “Which customer segment has the highest average balance?”

Outliers: extreme observations that deserve investigation

An outlier may be an error, a rare valid case, a new regime — or the most interesting observation in the dataset.

Mostly regular pattern



Data error

Extra zero, wrong sign, wrong unit, duplicate record.

Valid rare case

Very wealthy customer, large transaction, unusual product.

Process change

System migration, policy change, new branch behaviour.

Interesting signal

Potential fraud, stress event or operational exception.

Never delete an outlier just because it is inconvenient.

Detecting outliers with the IQR rule

The interquartile range is robust and works well as a first-pass rule.



IQR rule

Q1 = 25th percentile

Q3 = 75th percentile

$IQR = Q3 - Q1$

Flag values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$.

$Q1: =\text{QUARTILE.INC}(\text{range}, 1)$

$Q3: =\text{QUARTILE.INC}(\text{range}, 3)$

$IQR: =Q3 - Q1$

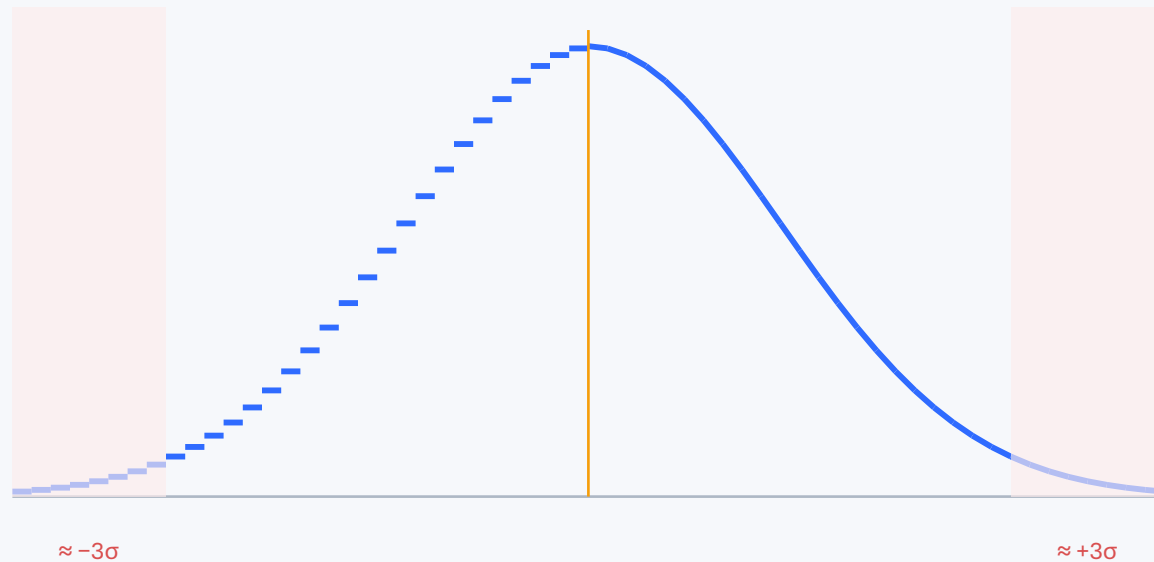
$\text{Lower} = Q1 - 1.5 \times IQR$

$\text{Upper} = Q3 + 1.5 \times IQR$

A statistical flag is a prompt to investigate, not a business verdict.

Detecting outliers with z-scores

A z-score expresses how many standard deviations an observation lies from the mean.



Formula

$$z = (x - \text{mean}) / \text{standard deviation}$$

```
= (B2 - AVERAGE ($B$2:$B$101)) / STDEV.S ($B$2:$B$101)
```

Caution

z-scores are sensitive to outliers and assume the mean/std are meaningful for the distribution.

For strongly skewed banking data, the IQR rule is often safer as a first screen.

Handling outliers: preserve information and business meaning

The right treatment depends on why the value is extreme.

Validate

Check source system, unit, sign, duplicate and business event.

Correct

Fix confirmed errors using traceable rules.

Keep

Valid extreme cases may be essential to risk analysis.

Transform

Log scale can reduce skew for positive amounts.

Cap / Winsorise

Limit extreme influence when methodologically justified.

Segment / model separately

Different populations may need different treatment.

Document every exclusion or transformation — especially in regulated analytical processes.

Missing values: “unknown” is a data state, not a number

Missingness can distort summaries and models — and may itself contain information.

x Missing values			
fx =COUNTBLANK (Table1[Income])			
Customer	Income	Balance	RiskGrade
C1001	65,000	12,500	A
C1002	54,000	8,200	B
C1003		45,500	A
C1004	45,000		C
C1005		165,000	

MCAR

Missing Completely At Random — unrelated to observed or unobserved data.

MAR

Missing At Random conditional on observed information.

MNAR

Missing Not At Random — missingness depends on the missing value or process.

Do not silently replace missing values with zero unless zero is the correct business meaning.

Detecting missing values in Excel

Combine formulas, filters and visual highlighting.

```
=COUNTBLANK (Table1[Income])
```

```
=ISBLANK (C2)
```

```
=COUNTIF (C2:C1000, "")
```

```
=IF (ISBLANK (C2) , "Missing" , "OK")
```

Be aware: a formula that returns "" may look blank but is not always the same as a truly empty cell.

x Exception list			
fx =IF (ISBLANK (B4) , "Missing" , "OK")			
Customer	Income	Balance	MissingFlag
C1001	65,000	12,500	OK
C1002	54,000	8,200	OK
C1003		45,500	Missing
C1004	45,000		Missing
C1005		165,000	Missing

Handling missing data: choose a strategy, do not hide the problem

Treatment depends on volume, mechanism, business meaning and downstream use.

Leave missing

Best when "unknown" is meaningful and downstream tools can handle it.

Exclude rows

Reasonable only when missingness is limited and exclusion does not bias the sample.

Impute simple values

Median for skewed numeric data; mode for categories — document the rule.

Create missing indicator

A separate flag can capture information in the fact that a value is missing.

Impute with a model

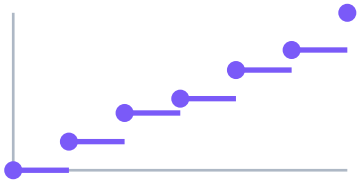
Useful for complex data, but adds assumptions and validation requirements.

Example simple rule: `=IF(ISBLANK(C2), MEDIAN($C$2:$C$1000), C2)`

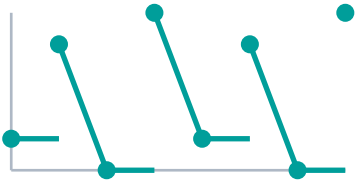
Patterns in data: what EDA is trying to discover

A pattern is a repeatable structure in the data that helps explain, predict or segment observations.

Trend



Seasonality



Clusters



Association



Anomaly



- Patterns are hypotheses until validated — they may reflect sampling, missingness or data quality problems.
- In banking, patterns often differ across segments, products, branches and time periods.
- A useful EDA habit: ask “what would make this pattern disappear?”

Pattern ≠ proof. It is a structured clue that guides the next question.

Data quality: six dimensions to check before trusting analysis

Quality is fitness for use: “good data” depends on the analytical purpose



The same dataset can be “good enough” for one purpose and unacceptable for another.

How data quality changes the analysis

Poor quality can create fake signals, hide real ones and bias decisions.

x Dirty data → distorted evidence		
fx =COUNTBLANK (Table1[Income])		
Issue	Example	Analytical effect
Missing	Income absent for some segments	Biased averages
Wrong type	Amounts imported as text	Rows silently excluded
Duplicate	Customer repeated twice	Inflated counts / balances
Outlier error	Balance ×1000	Mean & correlation distorted
Stale data	Old risk grade	Wrong segmentation



Wrong descriptive statistics

Mean, rate, dispersion and counts become misleading.

Wrong relationships

Spurious or suppressed correlations and segment differences.

Wrong decisions

Mis-prioritised customers, risk, operations or controls.

Data quality is part of the model – not a preprocessing footnote.

Excel EDA checklist: from raw table to trustworthy insight

A compact workflow to use after the course.

- 1 Define the question and the unit of analysis.
- 2 Convert the data to an Excel Table and preserve the raw extract.
- 3 Check types, units, keys and valid categories.
- 4 Profile rows, unique values, ranges and missingness.
- 5 Summarise numeric and categorical variables.
- 6 Visualise distributions, relationships and time patterns.
- 7 Investigate outliers and missing values before changing them.
- 8 Use correlation with scatter plots — not as proof of causality.
- 9 Assess data quality and document every transformation.
- 10 Communicate findings, limitations and next questions.

Good Data Science in Excel = clear question + clean structure + statistical thinking + visual evidence + data quality discipline